pynxtools-spm general table:

Parser type	Parser	Input format	Standard behaviour	Special behaviour	Formula / transformation	Notes
HDF5 vendor parsers	hfive_apex.py	.edaxh5	• Check HDF5 header + vendor fields (not the extension) • If supported: read header, phases, points with h5py → EbsdPointCloud • Write ROI overview + IPF into template • Missing data → warn and skip	• APEX: EBSD + EDS • Bruker, H5EBSD, EDAX, Oxford: EBSD (Oxford also microstructure) • DREAM.3D: filter-dependent content • EMsoft: detection only, writes nothing	• Euler: φ[rad] = φ[°]·π/180, φ<0 ⇒ φ + S • Grid: $x_i = i \cdot \Delta x$ (DREAM.3D $(i+½)\Delta x$) • EDAX: p = p_src or p_src·Δp (by version) • Lattice angles °→rad	• hfive_ebsd l.101: `supported == 2` never true → parser never used • hfive_ebsd l.274: `n_pts` may be undefined • `key=int` before `isdigit()` → ValueError • Failed stage still writes ROI output • Units/origin differ per vendor
	hfive_ebsd.py	.h5, .hdf5				
	hfive_edax.py					
	hfive_dreamthreed_legacy.py					
	hfive_emsoft.py	.h5				
	hfive_oxford.py	.h5, .h5oina				
Image parsers	image_png_protochips.py	.zip contianing .png	• Check TIFF header + vendor metadata (tag, XMP, XML or sidecar) • Flatten metadata → map with config tables • Each frame → greyscale array → flipped → image_2d + axes • No pixel size → unitless axes	• Vendor tag/sidecar differs: TFS, Zeiss, JEOL, TESCAN, Point Electronic, Hitachi, FEI • Hitachi needs TIFF + .txt • JEOL, TESCAN: sidecar only if no embedded metadata • Protochips: ZIP of PNGs → events	• Array = flipud(frame) • $x_i = i \cdot s$ (s = pixel size, else 1) • Hitachi: s = PixelSize[nm]·10^{-9} m • Zeiss: "K X" → ×1000	• TESCAN l.123–131: `txt` can be unbound • Hitachi l.124: IndexError on empty sidecar • Hitachi assumes nm • Any metadata ⇒ "supported" • Flip not applied to axes • Empty file → mmap error
	image_tiff_hitachi.py	.tif or .tiff + required .txt sidecar				
	image_tiff_jeol.py	.tif or .tiff, optionally plus .txt sidecar				
	image_tiff_tescan.py	.tif or .tiff, optionally plus .hdr sidecar				
	image_tiff_fei_legacy.py	.tif, .tiff				
	image_tiff_point_electronic.py					
	image_tiff_tfs.py					
	image_tiff_zeiss.py					
Nion parser	nxs_nion.py	.nsproj	• Index project files by UUID • Follow display items → .ndata / .h5 • Map metadata, then write data per item	• 6 on/off switches in `cfg` • Axis names reversed for generic data • sha256 only logged	• x□ = o + k·s • Unitless axes → int32 • Index attr = len − 1 − idx	• `verbose=True` default (l.66) • No path guard / empty-file crash • h5 attribute read before check (l.331) • Event id skipped on early return • `split("_")` vs `split(";")` (l.574)
RSCIIO-backed parsers	rsciio_bruker.py	.bcf	• Read whole file with rosettasciio • Keep valid objects • Per object: map metadata, write signal + axes	• Velox: version gate, hand-coded lenses • Gatan: dm2/3/4 • MSA: one spectrum • MRC: stack + sidecars (mdoc, rawtlt, shifts, xml) • Bruker: stub, no output	• x□ = o + k·s • Index attr = len − 1 − idx • MSA counts: uint64 or float64 • MRC: Å→m, kV→V, µm→nm	• MRC l.349: lone .mrc crashes • MRC l.654: wrong indentation • MSA l.156: broken path • Gatan l.125: event id +2 • Velox l.240: aperture index clash
	rsciio_gatan.py	.dm3, .dm4				
	rsciio_mrc.py	.mrc				
	rsciio_msa.py	.msa				
	rsciio_velox.py	.emd				
ELN/config parsers	oasis_config.py	*.oasis.specific.yaml *.oasis.specific.yml	• Match file name, load YAML • Map sections with config tables • List sections → numbered groups	• Context only, no measurement data • ELN: formula or atom types, ORCID • Config: description from first citation	• Counter: 1, 2, 3… per non-empty dict • Atom types: split → keep valid elements • ORCID → identifier	• ELN l.135: random element order • ORCID typed "DOI" (l.158) • Different user identifiers (cfg l.95 vs ELN) • Loose file-name test • Silent skips
	oasis_eln.py	eln_data.yaml				